Prototype Specification Document

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Interaction Mechanics & Input Paradigm Mapping

User Flow Selection – (Input Mechanics)

Check Grade

- Sign In
    - Enter Username and Password – (Point and click then Type or type in starting focus, tab navigate once and then type password)
    - Click the 'Sign in' Button – (Point and click, Click Enter while focus is in one of the above boxes)
    - Verify Using Duo – (Outside action, such as opening the app on another device)
- Choose Course – (Point and Click on title or card, Tab Navigate to correct Course)
- Find "Grades" on sideboard (Point and Click, Tab navigates; Might need to first open side panel by clicking on down arrow at the top of the screen.)
- Make sure filter is set correctly
    - Course should be the one you selected
    - Arrange By is usually by Due Date, set it as wanted.
- Move to assignment you want to check (Scroll, Drag, Tab Navigation)
    - Click on link to move to page (Point and Click, Tab Navigation and Enter)
        - On Page click on "View Feedback" to see comments from the teacher on the assignment if any have been left. (Point and Click, Tab Navigation and Enter)
        - To see other attempts select the attempt box (Point and Click, Tab Navigation and Enter)
        - To see review submission scroll to bottom of page. (Scroll, Drag, Tab Navigation)
            - Review comments on attempt (Scroll, Drag, Tab Navigation)
    - Test Alternate Grade by clicking existing score or pending score. (Point and Click, Tab Navigation and Enter)
- Scroll to bottom to check total class grade. (Scroll, Drag, Tab Navigation)

Fitts' Law Target Sizing

The smallest buttons are the sideboard links at between ~64px to ~200px width and a consistent 27px height in split screen and 36.8px in full screen. The goal is greater than or equal to 44X44px and the graded and pending scores with a 77.1x120px. This means that each button on the side bar is too thin to meet the minimum ergonomic size in split screen and Fullscreen.

High-Fidelity Prototyping & Micro-Interaction Design

Interactive Prototype

Was done using Figma here "https://www.figma.com/proto/Qq1eOKeBGvqxHT10JAscSD/Interactive-Prototype?node-id=5-245&t=qo6RFv6z7UniTjoX-1&scaling=scale-down&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=5%3A335 "

State Transition Definitions:

Dashboard

- Navbar

  All buttons use the same template

  - Default/Resting : White Box with name inside
  - Hover/Pointer-Over: Darkens when hovered over or pointed at.
  - Focus (visible, high-contrast focus indicator) Exists but doesn't allow for selection.
  - Active/Pressed: Resets the page or moves to the page.
  - Disabled (with accessible fallback explanations): None are disables
  - Loading/System Processing (providing immediate feedback for latency > 100ms): I have not experienced any latency.
  - Error/Validation Alert (inline error states with clear recovery guidance): There is a pop up that alerts if the button doesn't lead to an existing page.

    - Canvas: Takes you to Dashboard page
    - Profile: Pops an error message because that page is under construction
    - Canvas: Takes you to Dashboard page
    - Calendar: Pops an error message because that page is under construction

        - Inbox: Pops an error message because that page is under construction
        - History: Pops an error message because that page is under construction
        - Media: Pops an error message because that page is under construction
    - Dashboard Error
        - Goes away after 20 seconds
        - Goes away if clicked any ware that is not a functioning button.
    - Class Card
        - Moves to modules when pressed
        - Darkens when hovered over
        - Has three buttons
            - A – Pops error saying doesn't go anywhere yet, will be classes assignments
            - Q – Pops error saying doesn't go anywhere yet, will be classes quiz's
            - G – moves to classes grades

Modules

- Navbar – Same as above
- Sidebar
  All buttons use the same template
    - Default/Resting : Blue Underlined Text
    - Hover/Pointer-Over: Darkens when hovered over or pointed at.
    - Focus (visible, high-contrast focus indicator)
    - Active/Pressed: Resets the page or moves to the page or pops error.
    - Disabled (with accessible fallback explanations): None are disables
    - Loading/System Processing (providing immediate feedback for latency > 100ms): I have not experienced any latency.
    - Error/Validation Alert (inline error states with clear recovery guidance): There is a pop up that alerts if the button doesn't lead to an existing page.
- Modules
    - Title – Nonreactive
    - Row – Turns blue when hovered over
        - Pops error messages as assignments and quizzes are under construction.
    - Error Message – Disappears when clicked.

Grades

- Navbar – Same as above
- Sidebar – Same as above
- Course Dropdown
  Both function similarly
  - Default/Resting:
  - Hover/Pointer-Over: Darkens when hovered over or pointed at.
  - Focus (visible, high contrast focus indicator)
  - Active/Pressed: Resets the page or moves to the page or pops error.
  - Disabled (with accessible fallback explanations): None are disables
  - Loading/System Processing (providing immediate feedback for latency > 100ms): I have not experienced any latency.
  - Error/Validation Alert (inline error states with clear recovery guidance): I forgot to add it for this page
- Course – allows you to select other classes and load their versions of the lists, however there is only one class set up for now.
- Sort By – Changes how the graded assignments and quizzes are organized
  - Due Date: split them up by Date due with title bars separating each day
  - Assignment Group: shows only assignments
  - Quiz Group: shows only quizzes
  - Module: organizes them by modules displaying the module they are within.
- Assignments:
  - Always the same as it is primarily for display. This is mostly due to difficulty due to restrictions on the free version of Figma
  - Will allow for button drop downs for comparing against classmates, feedback for submitted assignment, and comments from teacher.

Accessibility Architecture & WCAG Specification

Semantic HTML Markup

Having it converted properly from Figma to HTML5 was an absolute nightmare, so I am unfortunately had to use AI to take my two Figma exports and work to convert it to a Html code. It is not truly dynamic as many components are hard coded.

Dashboard

The nav section details the various Global Pages. Canvas and Dashboard lead to dashboard page. The rest, Profile, Calendar, Inbox, History, and Media, all also pop the same message.

The header contains the name of the page.

The main contains a section that holds the class card, which includes a link to the module page for that class and three buttons that would go to assignments, quizzes, and grades. Due to Assignment and Quizzes Pages not existing I have popped up a message explaining that those are still under construction.

Modules

Navigation

This page still has the right sided global navigation bar, but also has a second bar next to it, the Course navigation bar, which I had made bigger than what canvas normally must meet accessibility standards. It allowed the user to move around the class focused resources, such as home and modules which currently lead to the same page, assignment and quiz which pop the error message, and finally grades which leads to the grade page.

Header

The Header contains a link to this page and defines who the user is, but due to being hard coded it is currently just says ‘User’.

Main

The main section of this is the module list which has the name of assignments and quizzes. In a fully functioning system clicking on them takes you to the appropriate page, but for now it just popped an error explaining that this part of the prototype is still under construction. It is also the due date for each.

Grades

Navigation

Same as the other two pages.

Header

Has the title “Class One > Grades > User”. ‘Class One’ links back to the module page, and ‘Grads’ links back to this page.

Main

The main encompasses a pair of labeled dropdown menus.
The first is meant to filter classes, as there is only one it shows only the one option already picked. If there were more changes it would change what class, you are currently in.

The Second is the is to filter the displayed grades. This is contained in a fieldset to allow for user interaction ease. There was the option for Due Date as default where the assignments and quizzes are separated by dated dividers, Assignment Group showing only assignments, Quiz Group showing only quizzes, Modules which organizes them into groups based on how they are presented in the modules page, and finally alphabetically.
The biggest change to how canvas normally does this is I have the filter put labeled dividers between sorted items, for example when filtering for due date the items are group by the day they are due. Or if it was module it takes the module name from Modules and groups them by what module they are in. I want this due to better viewing and ease of search.

ARIA & Screen Reader Hooks

With the help of AI in converting my Figma project pages into html, I was then able to add all appropriate hooks to the page. I was also able to make sure every button had a focus mode and was able to be navigated using alternative methods.

Dashboard
All components labeled appropriately, when necessary, such as the navbar being labeled with its current page. The error message has the appropriate labels and is marked as aria polite.

Modules

I made sure all each navbar had correct labels and navigation aids, such as aria current.
I added the aria-expanded="true/false" to the collapsible button meant to hide away the sideboard, modules title, and the collapse all button.

Each module has the correct pointers to understand the content within during navigation.

Grade

Once again, all navbars are labeled.

Each drop down has appropriate labels to allow for understanding of its purpose.

The table containing the grade items are all labeled, especially each unique divider.

And the error message is the same as the first two pages.

I verified that all that all colors I picked fall within WCAG 2.1 Aa standards.

Post-Mortem Reflections (Metacognition)

1. How did explicitly defining every component state (Hover, Focus, Active, Loading, Error) during the specification phase prevent dynamic usability bugs that static wireframes completely fail to address?

   It helped make planning and programing easier as I was able to set up a component that I could call on repeatedly if I needed to make any change to the pieces. It made sure I was thinking about how people would use my program.

2. Describe a scenario in your prototype where system latency or asynchronous data processing required specialized micro-interaction feedback (e.g., loading skeletons or aria-live announcements). How does your design keep the user informed and prevent repeated submission errors?

   A scenario where my prototype is them trying to access a page that doesn't exist. My design keeps the user informed using the pop up, explaining that those pages are not done. In the future I can have an animation, such as I had experimented with, that shows up to show a transition is occurring and give feed back to the user such

as a pop up saying “Processing Please Wait” especially if changing the filter on the grade takes a while.